By: Ana Júlia B. Bezerra

- **Characteristic Polynomial**

  The characteristic polynomial of a square matrix A is a polynomial that is essential in determining the eigenvalues of the matrix. For an $n \times n$ matrix A, it is defined as the determinant of the matrix $A - \lambda I$, where λ is a scalar (an eigenvalue) and I is the identity matrix of the same dimensions as A. Mathematically, this can be written as:

  $$p(\lambda) = \det(A - \lambda I)$$

  The characteristic equation is formed by setting the characteristic polynomial equal to zero, i.e.,

  $$\det(A - \lambda I) = 0$$

  Solving this equation gives the eigenvalues of the matrix A.

- **Determinant**

  The determinant of a square matrix A, denoted $\det(A)$, is a scalar value that provides important information about the matrix. For instance:

  - If $\det(A) = 0$, the matrix is singular and has no inverse.
  - The determinant of a matrix is also the product of its eigenvalues. For an $n \times n$ matrix with eigenvalues $\lambda_1, \lambda_2, \ldots, \lambda_n$, the determinant can be expressed as:

    $$\det(A) = \lambda_1 \cdot \lambda_2 \cdot \cdots \cdot \lambda_n$$

Problems

a) Calculate the characteristic polynomial of the matrix:

$$A = \begin{bmatrix} 1 & 2 & -1 \\ 1 & 0 & 1 \\ 4 & -4 & 5 \end{bmatrix}$$

1. Subtract λI from the matrix A

   The identity matrix I is:

   $$I = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

   Now subtract λI from matrix A:

   $$A - \lambda I = \begin{bmatrix} 1 & 2 & -1 \\ 1 & 0 & 1 \\ 4 & -4 & 5 \end{bmatrix} - \begin{bmatrix} \lambda & 0 & 0 \\ 0 & \lambda & 0 \\ 0 & 0 & \lambda \end{bmatrix} = \begin{bmatrix} 1-\lambda & 2 & -1 \\ 1 & -\lambda & 1 \\ 4 & -4 & 5-\lambda \end{bmatrix}$$

2. Find the determinant of $A - \lambda I$

   Now, compute the determinant of the matrix $A - \lambda I$:

   $$\det(A - \lambda I) = \det \begin{bmatrix} 1-\lambda & 2 & -1 \\ 1 & -\lambda & 1 \\ 4 & -4 & 5-\lambda \end{bmatrix}$$

   Use cofactor expansion along the first row. The formula for determinant expansion is:

   $$\det(A - \lambda I) = (1-\lambda)\det \begin{bmatrix} -\lambda & 1 \\ -4 & 5-\lambda \end{bmatrix} - 2\det \begin{bmatrix} 1 & 1 \\ 4 & 5-\lambda \end{bmatrix} + (-1)\det \begin{bmatrix} 1 & -\lambda \\ 4 & -4 \end{bmatrix}$$

3. Compute the 2×2 determinants

   - Compute the first 2×2 determinant:

     $$\det \begin{bmatrix} -\lambda & 1 \\ -4 & 5-\lambda \end{bmatrix} = (-\lambda)(5-\lambda) - (1)(-4) = -5\lambda + \lambda^2 + 4 = \lambda^2 - 5\lambda + 4$$

   - Compute the second 2×2 determinant:

     $$\det \begin{bmatrix} 1 & 1 \\ 4 & 5-\lambda \end{bmatrix} = (1)(5-\lambda) - (1)(4) = 5 - \lambda - 4 = 1 - \lambda$$

- Compute the third 2×2 determinant:

$$\det \begin{bmatrix} 1 & -\lambda \\ 4 & -4 \end{bmatrix} = (1)(-4) - (-\lambda)(4) = -4 + 4\lambda = 4\lambda - 4$$

4. Put it all together
   Substitute the 2×2 determinants into the cofactor expansion:

$$\det(A - \lambda I) = (1 - \lambda)(\lambda^2 - 5\lambda + 4) - 2(1 - \lambda) + (-1)(4\lambda - 4)$$

5. Simplify the expression
   First, expand $(1 - \lambda)(\lambda^2 - 5\lambda + 4)$:

$$(1 - \lambda)(\lambda^2 - 5\lambda + 4) = \lambda^2 - 5\lambda + 4 - \lambda^3 + 5\lambda^2 - 4\lambda = -\lambda^3 + 6\lambda^2 - 9\lambda + 4$$

   Now simplify the rest:

$$-2(1 - \lambda) = -2 + 2\lambda$$
$$-(4\lambda - 4) = -4\lambda + 4$$

   Finally, combine everything:

$$\det(A - \lambda I) = (-\lambda^3 + 6\lambda^2 - 9\lambda + 4) + (2\lambda - 2) + (-4\lambda + 4)$$
$$= -\lambda^3 + 6\lambda^2 - 11\lambda + 6$$

   The characteristic polynomial is:

$$p(\lambda) = \lambda^3 - 6\lambda^2 + 11\lambda - 6$$

b) A real matrix A (2×2) whose set of eigenvalues is $\{-2, 3\}$ has which determinant?

The determinant of a matrix is the product of its eigenvalues. Given the eigenvalues $\{-2, 3\}$:

$$\det(A) = (-2) \times 3 = -6$$

Thus, the determinant of matrix A is -6.

c) The determinant of a real matrix A of dimension 3×3, whose eigenvalues are given by the set $\{0, 2, 3\}$, is?

Again, the determinant is the product of the eigenvalues. Here, the eigenvalues are $\{0, 2, 3\}$:

$$\det(A) = 0 \times 2 \times 3 = 0$$

Since one of the eigenvalues is zero, the determinant of matrix A is 0. This indicates that A is a singular matrix and has no inverse.

Reference: Nicholson, W. K. (2006). Elementary Linear Algebra. ISBN 85-86804-92-4.